ORYX 理工 INSTITUTE



ADVANCED SAFETY MANAGEMENT

Level 3 Answer Key

Marking guide — instructor copy

Controlled document

Eight module examination keys and parallel-form guidance



Level 3 Answer Key

Oryx Safety Academy · Oryx Institute

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Marking guide and answer key

CONTROLLED DOCUMENT

Instructor copy — not for distribution to learners

This key covers the eight module examinations. Parallel forms substitute the starred items noted per module. Keep separately from learner papers. Assignment rubrics are administered per the assessment pack.

Module 1 — SHEQ management systems and ISO 45001 principles

#	Type	Mark	Answer / required elements
1	Short	1	Plan: context, leadership, planning. Do: support, operation. Check: performance evaluation. Act: improvement.
2	MCQ	1	B. Licence conditions, client standards, voluntary commitments, and the pending OSH Bill flagged as pending
3	Short	1	Paper-only worker consultation; unmeasured management review; change management triggered only by incidents.
4	MCQ	1	B. They persist through turnover and fashion by fixing ownership, competence, measurement and correction
5	Short	1	Any objective naming the change, owner, date and verification, e.g. driver committee elected at both depots by end-Q2; near-miss rate reviewed at each management review.
6	MCQ	1	B. A voluntary commercial decision; clients may write it into contracts
7	MCQ	1	B. Risk and effort
8	Short	1	Legal and other requirements on a live register, objectives, and resourced programmes to achieve them.

Module 2 — Construction safety requirements in depth

#	Type	Mark	Answer / required elements
1	MCQ	1	B. At design review
2	Short	1	Sequence; hazards per step; controls; competencies; hold points with named signatories.
3	MCQ	1	B. Engineered items: designed/selected by a competent person, checked before first load, inspected on a rhythm
4	Short	1	What does this detail force a worker's body to do at height, in a hole, or under a load; what would it look like if it forced nothing; what changes at cost if the hazard is removed here.
5	Short	1	Equipment certification/inspection; lift category; ground assessment; load; exclusion zone; competent supervision; weather limits.
6	MCQ	1	B. Induction, daily high-risk briefings, weekly inspections, permits, coordination meetings, handover documentation
7	Short	1	Authorship by the doers captures real sequence and hazards, and earns revision-with-a-pencil engagement instead of read-never filing.
8	MCQ	1	B. Prevention through design

Module 3 — Mining safety requirements

#	Type	Mark	Answer / required elements
1	Short	1	Cite Namibian law as law; SA MHSA/SANS as adopted benchmarks; mark Namibian mine-specific regulations as to-be-verified with the Ministry of Mines and Energy.
2	Short	1	Ground control; mobile equipment interaction; energy isolation; dust and ventilation (with the working-alone rule).
3	MCQ	1	B. It is the earliest drift signal before collapse
4	MCQ	1	B. Years to decades

#	Type	Mark	Answer / required elements
5	Short	1	Purpose, scope, definitions, hazard description, control standards, roles, training requirements, inspection and records, review trigger.
6	Short	1	Verify the measurement and reporting are still alive before treating the improvement as real.
7	MCQ	1	B. Beacons, headlights, radio logging, and give-way discipline at marked intersections with truck radio calls
8	MCQ	1	B. Prohibited without exception

Module 4 — Occupational hygiene and environmental monitoring

#	Type	Mark	Answer / required elements
1	Short	1	Anticipate, recognise, evaluate, control, re-verify.
2	MCQ	1	B. The machine's environment, not the worker's
3	Short	1	The time-weighted average integrates concentration across the full shift duration; twice the duration at the same concentration doubles the TWA against an 8-hour basis.
4	MCQ	1	B. Its source; limits differ by jurisdiction and list
5	Short	1	Any of: personal sampling in breathing zones; full-shift coverage of all exposed groups/shifts; OELs with named source; TWA treatment of extended shifts.
6	MCQ	1	B. Buy-quiet procurement, enclosure and maintenance
7	Short	1	They depend on correct selection, fit, condition and wearing every time; wet methods and extraction control the hazard at source for everyone.
8	Short	1	Engineering and administrative controls, evaluated and costed, with protection as documented interim only.

Module 5 — Contractor management

#	Type	Mark	Answer / required elements
1	MCQ	1	B. Verifiable safety evidence: policy with accountable person, sampled competence, incident history with explanations, certification, references
2	Short	1	Reporting nothing; incident-free histories without narrative are a warning sign, not a credential.
3	Short	1	Applicable rules/standards by status; stop-work right without penalty; permit and reporting interfaces; insurance and compensation responsibilities; consequences of failure.
4	MCQ	1	B. Cannot be delegated away; the client manages interfaces
5	Short	1	Whether the contractor's stated systems operate on the ground over time, not conditions at a moment.
6	MCQ	1	B. Using it is commercially and personally safe, demonstrated and recorded neutrally
7	Short	1	The day's hot work and lifts so trades do not create hazards for each other.
8	MCQ	1	B. The next prequalification shortlist

Module 6 — Root-cause investigation: ICAM fundamentals

#	Type	Mark	Answer / required elements
1	Short	1	Latent: a weakness planted earlier (defect list with no owner). Active: the triggering act (the reach-in to clear the jam).
2	Short	1	Organisational factors; task and environmental factors; individual actions; preconditions.
3	MCQ	1	B. What made this act reasonable to this person at this moment?
4	Short	1	The change, owner, date, cost class, and a verification method.

#	Type	Mark	Answer / required elements
5	Short	1	Stopping at blame; stopping at the first cause; stopping at paperwork.
6	MCQ	1	B. A defeated defence normalised by daily practice under production pressure
7	Short	1	The investigation's honesty, no naming, system-level fixes, was noticed by the workforce and repaid with trust.
8	MCQ	1	B. Rarely, and never as an excuse to stop asking about the system around the act

Module 7 — Auditing and inspection techniques

#	Type	Mark	Answer / required elements
1	Short	1	Inspection observes conditions at a moment; audit verifies a system against a standard via objective evidence; verification checks that stated arrangements actually operate.
2	Short	1	E.g. “Show me the training records for the two newest operators, with dates, content and sign-off, and who signed them.”
3	MCQ	1	B. Comparing documents, interviews and observation; any two in conflict is itself a finding
4	Short	1	Major: system failure allowing serious harm or statutory breach. Minor: isolated lapse in a working system. Observation: weakness inviting drift. Improvement opportunity: beyond compliance, worth doing.
5	Short	1	The requirement; the evidence; the gap.
6	Short	1	The centre is always prepared; the edges, night shifts, newest workers, forgotten records, reveal the system's true state.
7	MCQ	1	B. Audits that humiliate produce sites that hide, suppressing honest root causes
8	MCQ	1	B. A 15:45 permit close-out round, no cost

Module 8 — Training of trainers

#	Type	Mark	Answer / required elements
1	Short	1	Relevance, experience, practice, respect.
2	MCQ	1	B. Objectives, then checks, then the shortest content-and-practice path
3	Short	1	Owned with a commitment date, then met: “I will find out and send it by Friday”, and sent by Friday.
4	Short	1	Silence: small safe questions and visible action. Derailing: acknowledge, park, return if time. Cynicism: facts with respect, one convert at a time.
5	Short	1	It is not observable; objectives must state behaviour a check can prove, e.g. “each learner writes a four-part hazard report”.
6	MCQ	1	B. Assist, co-deliver, observed sign-off, credentials verified
7	Short	1	Teaching craft certified; the safety credential is still required for lead-instructor appointment.
8	MCQ	1	B. Day one taught workers the site was interested in what they noticed

Parallel forms

Each module's parallel form replaces questions 2, 4 and 7 of that module with the equivalents held in the secure parallel bank, and is administered under identical conditions after a minimum of one month. Parallel items are reviewed at each annual moderation.

Marking rules and records

One mark per question; no half marks; short answers earn the mark when listed elements appear regardless of order or wording. Pass per module: 6 of 8. Where a cohort's failure pattern clusters on one module, the finding is recorded in the course completion report and reviewed against the manual's teaching of that module. Results are entered on the Assessment Results Register (OSA-F10) on the day of marking.